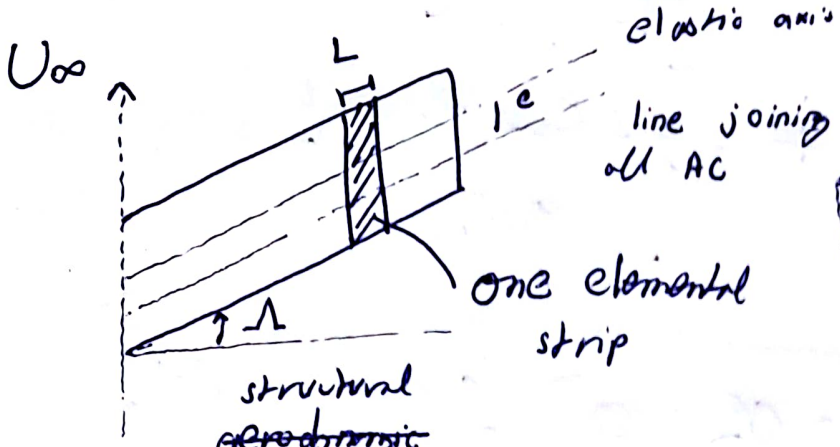
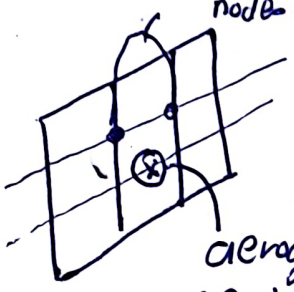
Divergence of a Swept Wing



Strip theory



every node has 3 DoF:

$$[\Delta y, \theta_z, \theta_x]$$

every element has 2 such nodes: 6 DoF

$$[\Delta y_1, \theta_{z_1}, \theta_{x_1}, \Delta y_2, \theta_{z_2}, \theta_{x_2}]$$

For every element we are interested in two parameters α, δ.

$(\delta = 0) \rightarrow$ no control surfaces.

$$\Delta\alpha = \frac{1}{2}\left[-\sin\Lambda\left(\theta_{z_1} + \theta_{z_2}\right) + \cos\Lambda\left(\theta_{x_1} + \theta_{x_2}\right)\right]$$

$$\boxed{\Delta\alpha = T_{\alpha s}\, U}$$

The only force applied is via aerodynamics.

For every element we have:

$$\begin{Bmatrix} L \\ m \end{Bmatrix}_e = q\, S\, AF_c \begin{Bmatrix} \alpha \\ \delta \end{Bmatrix}$$

M: about aerodynamic centre.

$$\begin{Bmatrix} L \\ m \end{Bmatrix}_e = qSC \begin{bmatrix} C_{l_\alpha} & C_{l_\beta} \\ 0 & (C_{l_\delta} \dots) \end{bmatrix} \begin{Bmatrix} \Delta\alpha \\ \Delta\delta \end{Bmatrix} + \begin{Bmatrix} qS\, C_{L_\alpha}\, \alpha_{rigid} \\ @M_{AC} \end{Bmatrix}$$

$C_{L_\delta} = 0$

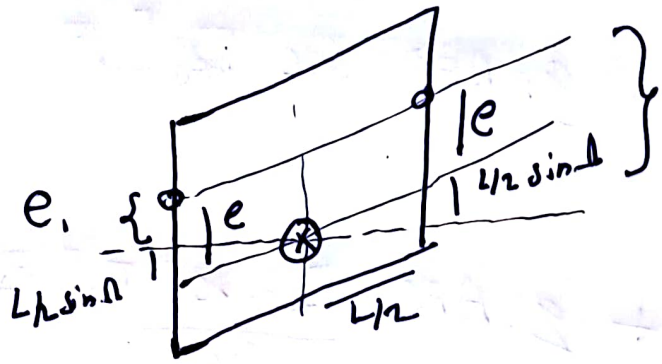
$$\begin{Bmatrix} L \\ m \end{Bmatrix}_e = qSC \begin{bmatrix} C_{l_\alpha} & 0 \\ 0 & 0 \end{bmatrix} \begin{Bmatrix} \alpha_{rigid} + \Delta\alpha \\ 0 \end{Bmatrix} + qSC \begin{Bmatrix} 0 \\ C_{m_{AC}} \end{Bmatrix}$$

$\$$ ($C_{m_{AC}} = 0$ for sym airfoil)

The aerodynamic force creates forces on the structural nodes of the form:

$$F = (T_{AS})^T \{F\}_e$$

$$T_{AS} = \begin{bmatrix} 1 & -e_1 \sin\Lambda & e\cos\Lambda & 1 & -e_2 \sin\Lambda & e_2\cos\Lambda \\ 0 & -\sin\Lambda & \cos\Lambda & 0 & -\sin\Lambda & \cos\Lambda \end{bmatrix}$$



$$e_1 = e - \frac{L}{2}\sin\Omega$$
$$e_2 = e + \frac{L}{2}\cos\Omega$$

also, for each structural node.

$$F = K U$$ U: displacement node.

where $[K]_{n\times n}$ is a bandel matrix formed by elemental $[K_E]_{6\times 6}$

Now:

$$F_A = q S C_{L_2} C(\alpha + \Delta\alpha)$$

$$F_A = q S C_{L_2} C\, \alpha_{rigid} + q S C_{L_2} C\, T_{2S}\, U$$

$$F_A = q S C_{L_2} C\, \alpha_{rigid} + q S C_{L_2} C\, T_{2S} K^{-1} F$$

replace C_{L_2} by AFS

$$AFS = S_C \begin{bmatrix} C_{L_2} & 0 \\ 0 & 0 \end{bmatrix}$$

$$F = T_{AS}^{\Gamma} F_A$$

~~$F_A = q \cdot AFS \begin{Bmatrix} \lambda_{rigid} \\ 0 \end{Bmatrix} + q \cdot AFS \cdot SIC \, T_{AS}^T F$~~

$$F_A = q \cdot AFS \begin{Bmatrix} \lambda_{rigid} \\ 0 \end{Bmatrix} + q \cdot AFS \underbrace{T_{LS} \cdot K^{-1} T_{AS}^T F_A}_{SIC^{-1}}$$

$$\Rightarrow F_A (I - q \cdot AFS \cdot SIC) = q \cdot AFS \begin{Bmatrix} \lambda_{rigid} \\ 0 \end{Bmatrix}$$

if $(I - q \cdot AFS \cdot SIC) \rightarrow$ becomes singular,

then divergence $\Rightarrow F_A$ blows up.

let λ_{MAX} = max. eigen value $(AFS.SIC)$

$$\boxed{q_D = \frac{1}{\lambda_{MAX}}}$$ — divergence speed.